

Symmetric lens

`MathlibAnnex.symmetricLens`

Names the reflection-symmetric bounded region used to force preservation of a midpoint.

Statement

``symmetricLens x y r`` is the intersection of the two closed balls of radius r centered at x and y .

Definition

A point lies in the symmetric lens exactly when its distance from each focus is at most r . The common midpoint is the center of the point-reflection symmetry that exchanges the two foci.

Assumptions

A normed additive commutative group E ; points x and y ; a real radius r .

Conclusion

The symmetric lens is the intersection of the closed radius- r balls centered at x and y . This definition needs no positivity assumption.

Main citations

Lean source signature (exact)

```
def symmetricLens (x y : E) (r : ℝ) : Set E
```

[Read exact MathlibAnnex v0.4.0 source](#)

Exact Card identity

Stable Card ID: `b6295e62b69808246ec83346365d7b0e00269f627c02dcd6ce3d253fb920949a`

Card SHA-256: `3935024dd76f1c0d54f0ff1aa13a35e31d2a4b7ebce1fae22ae01e793e6eba20`

Approved exposition SHA-256: `88e659789bb029e6bbfef033cf263f4258b519e3058ebc9910cd86e2021b7197`

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